

SOFTWARE ENGINEERING

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS14	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

1. To familiarize with basic Software engineering methods and practices, and its applications.
2. To explain layered technology in software engineering
3. To teach software metrics and software risks.
4. To familiarize with software requirements and the SRS documents.
5. To facilitate students in software design

COURSE OUTCOMES

1. Analyze software development life cycle and select appropriate model suited for diverse software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document
3. Conceptualize the system through design with emphases on architectural modeling and user interface
4. Classify software testing strategies and recommend testing techniques during the construction of software.
5. Examine the application of metrics and software tools during software development

UNIT-I

INTRODUCTION TO SOFTWARE ENGINEERING

CLASSES: 14

INTRODUCTION TO SOFTWARE ENGINEERING: The Evolving nature of software engineering, Changing nature of software engineering, Software engineering Layers, The Software Processes, Software Myths.

PROCESS MODELS: A Generic Process Model, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Spiral Model, The Unified Process, Personal and Team Process Models, the Capability Maturity Model Integration (CMMI).

UNIT-II

REQUIREMENTS ENGINEERING

CLASSES: 14

REQUIREMENTS ENGINEERING: Functional and Non-Functional Requirements, The Software requirements Document, Requirements Specification, requirements Engineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management, System Modeling: Context Models, Interaction Models, Structural Models, Behavioral Model, Model-Driven Engineering.

DESIGN CONCEPTS: The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles.

UNIT-III

DESIGN AND IMPLEMENTATION:

CLASSES: 12

DESIGN AND IMPLEMENTATION: Design Patterns, Implementation Issues, Open Source Development. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation.

SOFTWARE TESTING STRATEGIES: A Strategic approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing.

UNIT-IV	PRODUCT METRICS	CLASSES: 10
PRODUCT METRICS: A Frame Work for Product Metrics, Metrics for the Requirements Model, Metrics for DesignModel, Metrics for Source Code, Metrics for Testing. PROCESS AND PROJECT METRICES: Metrics in the Process and Project Domains, Software Measurements, Metrics forSoftware Quality, Risk Management: Risk verses Proactive Risk Strategies, Software Risks, Risk Identification, Risk Projection, Risk Refinements, Risk Mitigation Monitoring and Management (RMMM), The RMMM Plan.		
UNIT-V	OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT	CLASSES: 10
OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT: Overview of SEI -CMM, ISO 9000, CMMI, PCMM, TQM and Six Sigma. OVERVIEW OF CASE TOOLS: Software tools and environments: Programming environments; Project management tools; Requirements analysis and design modeling tools; testing tools; Configuration management tools;		
TEXT BOOKS		
1. Software Engineering, A Practitioner's approach, Roger S. Pressman (2011), 7th edition, McGraw Hill International Edition, New Delhi. 2. Software Engineering, Sommerville (2001), 9th edition, Pearson education, India.		
REFERENCE BOOKS		
1. Software Engineering, K. K. Agarval, Yogesh Singh (2007), 3rd edition, New Age International Publishers, India. 2. Software Engineering an Engineering approach, Lames F. Peters, Witold Pedrycz(2000), John Wiely& Sons, New Delhi, India. 3. Systems Analysis and Design, Shely Cashman Rosenblatt (2006), 6th edition, Thomson Publications, India.		
WEB LINKS		
1. https://nptel.ac.in/courses/106/105/106105087/		